QP CODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION

FIFTH SEMESTER

MG5DSECFS302 - EMBEDDED SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer any **10** questions
Each question carries **2** marks

1. Identify the meaning of an embedded system. [U] / [CO1]
2. Distinguish between standalone and networked embedded systems. [U] / [CO1]
3. Differentiate between CISC and RISC processors. [U] / [CO1]
4. Describe what a System-on-Chip (SoC) is. [U] / [CO1]
5. Compare USB Type-C and Micro-USB in terms of connectivity and usage. [U] / [CO2]
6. Identify any two applications of USART. [U] / [CO2]
7. Discuss the applications of I²C in embedded systems. [U] / [CO2]
8. Analyse how two operating system functions contribute to the efficient management of computer resourses. [An] / [CO3]
9. Examine the role of a timer in managing process execution in an operating system. [An] / [CO3]
10. Explain the primary differences between the *Ready*, *Running*, and *Blocked* states of a task in an RTOS kernel. What specific event causes a task to transition from *Running* to *Blocked*? [U] / [CO3]

11.	Show how a telephone answering machine records a caller's message.	[An] / [CO3]
12.	A 100 KB data file is reduced by 20% after compression. Calculate its new size.	[A] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13.	Discuss the differences between general purpose processors and DSP processors with examples.	[U] / [CO1]
14.	Discuss the role of debugging and testing tools in embedded systems.	[U] / [CO1]
15.	Describe the role of a Wireless Access Point (WAP) in a Wi-Fi network. Explain how it connects wireless devices to a network.	[U] / [CO2]
16.	Classify the common types of Bluetooth devices based on their applications and explain each category with examples.	[U] / [CO2]
17.	Explain the concept of timer functions in an operating system.	[U] /[CO3]
18.	Illustrate with a brief explanation how a message queue facilitates data passing between an Interrupt Service Routine (ISR) and a processing task.	[An] /[CO3]
19.	Use an ECU to show how sensor information is converted into an engine control action.	[A] / [CO4]
20.	A device produces a large amount of data but has limited memory. Use data compression to show how the problem can be reduced.	[A] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

21.	Describe the software life cycle and embedded life cycle, and summarize the role of modeling in embedded systems.	[U] / [CO1]
22.	Discuss the purpose of an RTOS (Real-Time Operating System) kernel?	[U] / [CO3]
23.	Use embedded-system concepts to show the main operations of a digital camera from capturing an image to storing the photograph.	[A] / [CO4]

[10x2 = 20]